## POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024

Programme: BE | Full Marks: 100

Course: Applied Mathematics | Pass Marks: 45

Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*
*The figures in the margin indicate full marks.*
*Attempt all the questions.*

1. a) Define analyticity of a function f(z). Show that the necessary condition for the function $f(z) = u(x,y) + iv(x,y)$ to be analytic on a domain $D$ is $u_x = v_y$ and $u_y = -v_x$ at each point $(x,y)$ of $D$. 7

   b) State and prove Cauchy integral formula. Evaluate the integral $\oint_c \frac{e^{5z}}{(z+i)^4} dz$, where c: $|z| = 2$. 8

   **OR**

   State Cauchy's residue theorem and using it, evaluate: $\oint_c \frac{2z}{(z+1)(z-1)^3(z+3)} dz$ where c:$|z|$ =2 counter clockwise.

2. a) Find the expansion of $\frac{7z-2}{z(z+1)(z-2)}$ in the region given by 7

   i. $0< |z+1| < 1$.
   ii. $1 < |z+1| < 3$.

   b) Define bilinear transformation. Find the bilinear transformation which maps the points $z = 0, -1, i$ onto the points $w = i, 0, \infty$. Also, find the image of the unit circle $|z| = 1$. 8

3. a) State and prove first shifting theorem on Z-transform. Find Z-transform of $e^{\frac{in\pi}{2}}$ and then find $Z(cos\frac{n\pi}{2})$ and $Z(sin\frac{n\pi}{2})$. 7

   b) Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 2n$, $y_0 = 0$, $y_1 = 0$ by using Z-transform. 8

4. a) Show that $Z[nf(t)] = -z\frac{d}{dz}[F(z),]$ where $F(z) = Z[f(t)]$. 7

   Find $Z^{-1}\left[\frac{z}{(z+1)^2(z-1)}\right]$.

   b) Find the solution of one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$ with initial velocity $g(x)$, initial deflection $f(x)$ and boundary condition u(0, L) = 0 = u(L, t). 8



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1. a) Define analyticity of a function f(z). Show that the necessary condition for the function $f(z) = u(x,y) + iv(x,y)$ to be analytic on a domain $D$ is $u_x = v_y$ and $u_y = -v_x$ at each point $(x,y)$ of $D$. 7

   b) State and prove Cauchy integral formula. Evaluate the integral $\oint_c \frac{e^{5z}}{(z+i)^4} dz$, where c: $|z| = 2$. 8

2. a) State Cauchy Residue Theorem. By applying Cauchy Residue Theorem, evaluate $\oint_C \frac{dz}{(z^2+4)^3}$ where C : $|z - i| = 2$. 8

   b) Find the linear transformation that maps z = 0, z = 1, z = ∞ into the points w = – 3, w = – 1, w = 1 and find the fixed point of the transformation. 7

3. a) State and prove first shifting theorem on Z-transform. Find Z-transform of $e^{\frac{in\pi}{2}}$ and then find $Z(cos\frac{n\pi}{2})$ and $Z(sin\frac{n\pi}{2})$. 7

   b) Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 2n$, $y_0 = 0$, $y_1 = 0$ by using Z-transform. 8

4. a) Show that $Z[nf(t)] = -z\frac{d}{dz}[F(z),]$ where $F(z) = Z[f(t)]$. 7

   Find $Z^{-1}\left[\frac{z}{(z+1)^2(z-1)}\right]$.

   b) Derive one dimensional heat equation. 8

5. a) Change the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar form $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$. 8

   b) The diameter of a semi-circular plate of radius 'a' is kept at 0°C and the temperature at the semi-circular boundary is T°C. Find the steady state temperature of the plate. 7

OR

Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature

is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$.

6. a) Find the Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$, and then show that $\int_0^\infty \frac{\cos kx}{1+x^2}dx = \frac{\pi}{2}e^{-k}$. 7

OR

Find the Fourier transform of $f(x) = xe^{-x^2}$.

b) Show that $\int_0^\infty \frac{\cos wx + w\sin wx}{1+w^2}dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$. 8

7. Attempt **any two** questions. 2×5

a) Find Laurent series of the function $f(z) = \frac{z+3}{z(z+1)(z-2)}$ in the region $1 < |z| < 2$.

b) Find the solution of the differential equation, $y^2u_x - x^2u_y = 0$, by using separating of variables.

c) Show that the function $u(x, y) = e^{2x}(x\cos 2y - y\sin 2y)$ is a harmonic function. Find its harmonic conjugate.



5. a) Express the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar coordinates. 7

OR

Derive two-dimensional heat equation completely with necessary assumptions.

b) Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature 8

is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$.

6. a) Find the Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$, and then show that $\int_0^\infty \frac{\cos kx}{1+x^2}dx = \frac{\pi}{2}e^{-k}$. 7

b) Show that $\int_0^\infty \frac{\cos wx + w\sin wx}{1+w^2}dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$. 8

7. Attempt all the questions: 4×2.5

a) Check analyticity of $f(z) = z^2$.

b) Find $Z(a^n)$.

c) Find the solution of the partial differential equation $u_{xx} + 9u = 0$.

d) Define linear partial differential equation with suitable example.

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*Candidates are required to give their answers in their own words as far as practicable.*

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***Attempt all the questions.***

1. a) Define analytic function. Show that the necessary condition for a function $f(z) = u + iv$ to be analytic is $u_x = v_y$ and $u_y = -v_x$. 7

   b) Define harmonic function. Check $u = \sin x \cosh y$ is harmonic or not? If yes, find a corresponding harmonic conjugate $v$ of $u$ and corresponding analytic function $f(z)$. 8

2. a) Find the Laurent's series expansion of $f(z) = \frac{4z+3}{z^3-z^2-6z}$ in the following regions 7

   i. $0 < |z| < 2$

   ii. $2 < |z| < 3$

   iii. $|z| > 3$

   **OR**

   Define bilinear transformation. Find the bilinear transformation which maps the points $z = 0, -1, i$ onto the points $w = i, 0, \infty$. Also, find the image of unit circle $|z| = 1$.

   b) State and prove Cauchy integral formula. Evaluate: $\oint_C^{\square} \frac{e^{2z}}{(z-1)^4}\, dz$, where $C$ is the circle $|z| = 2$ in counter clockwise direction. 8

3. a) Find:

   i. $Z^{-1}\left[\frac{z}{(z^2-5z+6)}\right]$ 3

   ii. $Z^{-1}\left[\frac{z}{(z+2)(z-1)^2}\right]$ 4

   b) Define Z-transform of a function. Prove that 8

   $Z[a^n f(t)] = F\left(\frac{z}{a}\right)$, where $Z[f(t)] = F(z)$.

   Using it, find $Z(a^n e^{ibt})$ and hence deduce the values of $Z(a^n \cos bt)$ and $Z(a^n \sin bt)$.



4. a) Using Z-transform, solve the difference equation: 7

   $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$, $y_0 = 0$, $y_1 = 0$

   b) Derive one dimensional wave equation with required assumptions. 8

   **OR**

   Express the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar co-ordinates.

5. a) A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is 7

   $$f(x) = \begin{cases} x & \text{if } 0 \le x \le 50 \\ (100-x) & \text{if } 50 < x \le 100 \end{cases}$$

   Find the temperature $u(x,t)$.

   b) Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ assuming the appropriate initial and boundary conditions. 8

6. a) Using Fourier integral representation, show that 7

   $$\int_0^{\infty} \frac{\cos xw + w \sin xw}{1+w^2}\, dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0. \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

   b) Find the Fourier cosine transform of $f(x) = e^{-x}$ and by using Parseval's identity, show that $\int_0^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$. 8

7. Attempt all the questions 4×2.5

   a) Evaluate: $\oint_C^{\square} \frac{2z}{(z-1)(z+3)} dz$, where $C$ is the circle $C: |z| = 2$ in counter clockwise direction by using Cauchy residue theorem.

   b) Check whether the function $u(x,y) = z\bar{z}$ is analytic or not.

   c) Find $Z(n^2)$.

   d) Solve $u_x + u_y = 0$ by using separation of variables.

## POKHARA UNIVERSITY

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Course: Applied Mathematics | | Pass Marks : 45
| | Time : 3 hrs.

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***Attempt all the questions.***

1. a) State and prove Cauchy-Reimann equation as necessary conditions for function to be analytic. 5

   b) Prove that u $=y^3 - 3x^2y$ is a harmonic function. determine its harmonic conjugate and find the corresponding analytic function f(z) = u + iv. 5

   c) State Cauchy Integral formula for derivative. Evaluate: 5

   $\oint_c \frac{z^6}{(2z-1)^6}$ dz, where c is the unit circle $|z|=1$, counterclockwise.

2. a) Find the series expansion of the function $f(z)=\frac{1}{z^2-3z+2}$ in the regions: 5

   i. $|z|<1$

   ii. $1<|z|<2$

   b) Define singularity, zeros, and poles of a function. Evaluate $\oint_c f(z)dz$ where f(z) = $\frac{e^{2z}}{(z+1)^3}$ where c is the ellipse $4x^2+9y^2=16$. 5

   c) Define bilinear transformation. Find the bilinear transformation which makes the point z = 0, -1, i onto w = i, 0, ∞. Also find the image of unit circle $|z| = 1$. 5

3. a) Define Z-transform of a function f(n). Find the Z-transform of $e^{in\pi/2}$ and hence find $Z[\cos(\frac{n\pi}{2})]$ and $Z[\sin(\frac{n\pi}{2})]$. 8

   b) Solve $U_{n+2} - 2\cos\alpha\, U_{n+1} + U_n = 0$ where, by using z-transform. 7



4. a) Using Fourier integral representation, show that 7

$$\int_0^\infty \frac{\cos xw + w\sin xw}{1+w^2}\,dw = 0 \ \text{if } x<0$$
$$= \frac{\pi}{2} \ \text{if } x=0$$
$$= \pi e^{-x} \ \text{if } x>0$$

   b) Find the Fourier cosine transform of f(x) = $e^{-mx}$ for m > 0 and then Show that $\int_0^\infty \frac{\cos kx}{1+x^2}dx = \frac{\pi}{2}e^{-k}$. 8

5. a) Solve one-dimensional wave equation with initial deflection is $0.01\sin 3x$ and initial velocity is zero and $L=\pi$, $c^2 = 1$. 7

   b) Determine the solution of one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$ Subject to the boundary condition u(0, t) = u(L, t) = 0 and initial condition is u(x,0) = $\begin{cases} -x, & 0<x\le\frac{L}{2} \\ L-x, & \frac{L}{2}\le x<L \end{cases}$. 8

   where L being the length of the bar.

**OR**

   Derive two -dimensional heat equation with required assumptions.

6. a) When two- dimensional Heat equation $\frac{\partial u}{\partial t} = c^2(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2})$ become Laplace equation. By the concept of solution of Laplace equation with rectangular boundaries, solve the problem: 7

   A rectangular plate with insulated surface is 8cm wide so long compared to its width that it may be considered infinite in length without introducing an appreciable. If the temperature along the short edge y=o is given by u(x,0) =100 $\sin\frac{\pi x}{8}$, $0<x<8$ while two long edges x=0 and x=8 as well as the other short edge are kept at $0^o$c. Show that steady state temperature at any point of the plate is given by u(x, y) = $100e^{-\pi y/8}\sin\frac{\pi x}{8}$.

**OR**

   Derive polar form of Laplace equation.

   b) Find the deflection u(x,y,t) of a square membrane with a = b = 1 with c = 1, if the initial velocity is zero and the initial deflection is $0.1\sin 3\pi x\ \sin 4\pi y$. 8



7. Solve: (Ab...

a) Find...

7. Solve: (Any two) 2×5

a) Find the z-transform of $Z(a^n cosbt)$.

b) Show that $\int_0^\infty \frac{w\sin xw}{a^2+w^2}dw = \frac{\pi}{2}e^{-ax}$ $where \quad x>0, a>0$.

c) Solve $u_x + u_y = 0$ by using sepation of variables.